JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR FRONTIER MODELS IN AN OPEN-WORLD RPG

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ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, which places a frontier model inside Heroes of Jin Yong, a 1996 open-world role-playing game run under DOS emulation. The ending the game defines, recovering the fourteen books of the Jin Yong novels, takes a first-time human player tens of hours. The game asks for perception of 320×200 isometric pixel art and 16-pixel Traditional Chinese, knowledge of the novels its world is built from, and a keyboard as the only tool. It asks for a map the model keeps for itself, turn-based fights against the AI of the game, and one plan across all of it, with no quest log, marker or score. Progress is scored as milestones of the game itself, and by completion time once a model finishes. Under a 60-minute budget, one of 12 frontier models plays past the opening of the game, and none gains experience or holds a book. The benchmark, replays and every session are public at `https://github.com/hanxiao/jy-crpg-bench`.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and the benchmarks that measure it have so far run in text, as business simulations (Backlund & Petersson, 2025; Fan et al., 2026) or interactive fiction (Phan et al., 2025). A game adds the requirement they leave out: the model has to recover its own state from raw perception, because nothing in the environment reports it. Game benchmarks for these models each remove one part of that problem, whether perception, language, cultural prior or game knowledge (Paglieri et al., 2025; ARC Prize Foundation, 2026; Ju et al., 2026), so the combination stays untested.

We present JY-CRPG-BENCH, built on Heroes of Jin Yong (金庸群俠傳), a 1996 Chinese role-playing game that we run as its original DOS binary under emulation. Its ending requires recovering the fourteen books of the Jin Yong novels from an open world, a task that takes a first-time human player tens of hours. The game asks for much of what a frontier model is expected to have, all at once, and Figure 1 shows that world. The model must read 320×200 pixel art drawn in a $45°$ isometric projection, with Traditional Chinese in 16-pixel glyphs and no key that moves straight up the screen. It must bring knowledge the game never states. The world is assembled from the fourteen novels, so who a character is, which faction to join and which storyline a book sits behind are known to a reader of the novels and written nowhere on screen. Its only actuator is a keyboard, reached through an API as a tool. The game keeps no quest log, draws no marker and reports no score, so the model must keep its own model of the world it walks and decide for itself what is worth doing. ARC-AGI-3 places the same goal-setting requirement at the centre of agentic intelligence (ARC Prize Foundation, 2026). Its fights are turn-based on a grid against the AI of the game, with position, range and inner force to manage. All of it is one long-horizon task: a persistent world with sparse and delayed reward over thousands of dependent steps. That is the setting reinforcement learning was built for, and here a frontier model plays it in context. Under a 60-minute budget, one of 12 frontier models plays past the opening, and even it stands at its first fight with the fourteen books still to find.

(a) the scene every session starts in (b) the overworld the scenes hang off (c) the four axes the game moves along (d) dialogue, 16-pixel glyphs (e) the menu inside a scene (f) the record the benchmark reads (g) the bag, one item described at a time (h) the outdoor menu and the party (i) a turn of battle on the grid

Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four movement axes (c), the three kinds of text panel (d–f), the bag (g), the outdoor menu with the party list (h), and a turn of battle (i).

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. The next section reviews game environments as agent benchmarks and the measurement problem that long horizons create. Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports the field: how many milestones are open, where they separate the models, and how the budget is spent.

Table 1: Interactive agent benchmarks. The observation column is what the model receives before any upscaling; a self-directed environment supplies no task list, score or objective marker during play; the last column is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
JY-CRPG-BENCH	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save, replay
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026); the opening of our environment has that structure.

The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2022), and NetHack remains the deepest environment in use for learned agents (Küttler et al., 2020).

2.2 Long-Horizon Evaluation and Its Measurement

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the setting to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), and TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025). ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Long horizons strain measurement. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025). An audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024), and our measurement follows the same principle.

3 JY-CRPG-Bench

3.1 Environment

Heroes of Jin Yong, published by Heluo Studio (河洛工作室) in 1996, opens with the protagonist in a small compound. The fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor world map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Every scene carries an entry condition. Of the 84 scenes, 7 are open at the start: the home, five inns (客棧) and the house of the hermit (南賢). The conversation with the hermit opens all but 4 of the rest, and from there the model chooses the order of play.

The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit. A cabinet beside him holds the compass (羅盤), the only source of coordinates the game offers, and his conversation opens the house of the first companion, who joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force (內力), and the first scripted fight opens by draining it. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one; some of these fights end the game when lost, which then offers nothing but its own load menu, and others let the story continue.

Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, a script that multimodal evaluation has largely neglected (Tam et al., 2025). The character sheet on screen displays the same record the benchmark reads from memory. Two further properties recommended the title. It is menu-driven, so a model that deliberates between actions is not disadvantaged as it would be in a real-time game (cf. Zhang et al., 2025a), and its progression state is rich enough to measure at the level of individual character statistics.

3.2 Observation and Action Space

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The frame sits at the hard end of several documented weaknesses of vision-language models. Downsampling costs them accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), and images far from the training data degrade recognition even of familiar categories (Lin et al., 2026). Text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads movement from the scrolling background loses count of its tiles there. Canopies and roofs are drawn over the character.

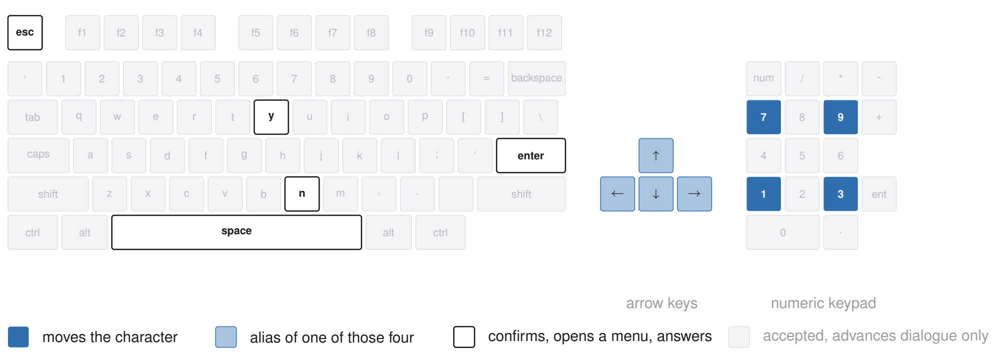

Figure 2: The keyboard the API exposes. The nine keys the game answers outside dialogue are picked out, the arrow keys are aliases of the four movement keys, and any key advances a dialogue.

The one action presses a key, or a list of keys in order, holding each for a requested number of emulated frames, ten by default. It returns once the screen has reacted and held still, waiting through a scene transition if one starts, so the frame a model receives is the one to act on. There is no call that advances the game without input, since the game only moves on input, and the hold is counted in frames because the game samples its keyboard once per game-loop iteration. A scored session answers four of the calls in Table 2: the screen, the brief, the key names and the action. Three further calls save, list and restore emulator snapshots for development, and a scored session refuses them, so a run cannot be rewound from outside the game. The save and load menu of the game stays open to the model as to a human player, so a model that loses a fight can reload the last save the game wrote, and the recording shows it doing so.

Table 2: The control surface. A scored session answers the four calls marked; the three snapshot calls serve development.

Call	Body	Meaning	Scored
`GET /api/screen`	–	the current frame	✓
`GET /api/help`	–	the brief, in English or Chinese	✓
`GET /api/keys`	–	every accepted key name	✓
`GET /api/slots`	–	emulator snapshots on disk	
`POST /api/key`	`{"key":"kp3","hold":10}`	press one key, or several in order	✓
`POST /api/save`	`{"name":"..."}`	take a snapshot	
`POST /api/load`	`{"name":"..."}`	restore a snapshot	

3.3 Harness and Session Protocol

The benchmark supplies the brief and the control surface, and the model plays through an agent harness. Every model in this paper ran inside pi[1], a minimal coding agent installed as shipped, whose four tools read, write and edit files and run a shell, with no web access, so no model could look up a walkthrough. The model drives the control surface with HTTP calls from that shell. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game. It is served in English and Chinese with the on-screen Traditional Chinese terms kept verbatim, so the language a model reasons in is not itself a handicap. An action returns only its metadata, so the model looks at the screen when it decides to, and the benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game. Any memory a model keeps between actions it writes for itself in that shell. The leaderboard is therefore a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b). The brief is published as `agents.md`, the project file pi reads at launch, and a scored session is refused every field that would report its own progress until the run ends, so a model cannot query its own score.

A session is created under a declared name with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. The wall clock of every decision call is kept on the server, so played time and think time are measured for any client and shown beside a run without ranking it. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix B, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or the game is completed, and there is no inactivity cutoff. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline, so the behaviour behind a score can be reviewed. Figure 3 shows the apparatus, and Appendix A lists the emulation and hosting parameters.

3.4 Measuring Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the archives the release ships. We locate the same array in the live emulated machine by anchoring on one character name and confirming the two neighbouring records at the expected stride, since the name recurs in the image and the layout moves between runs. The bag in front of that array is the working copy and changes the moment an item is taken. The item count, the compass and the books are read from it after every action and again from every save the game writes. A milestone, once seen, is kept, so an item picked up and later spent still credits it. Reading is passive: the benchmark serialises the machine and never loads state into it. The only keys it presses on its own are those of the save described next, which take the game a few seconds of menu time that is not charged to the run and leave the screen as they found it.

[1] `https://github.com/badlogic/pi-mono`

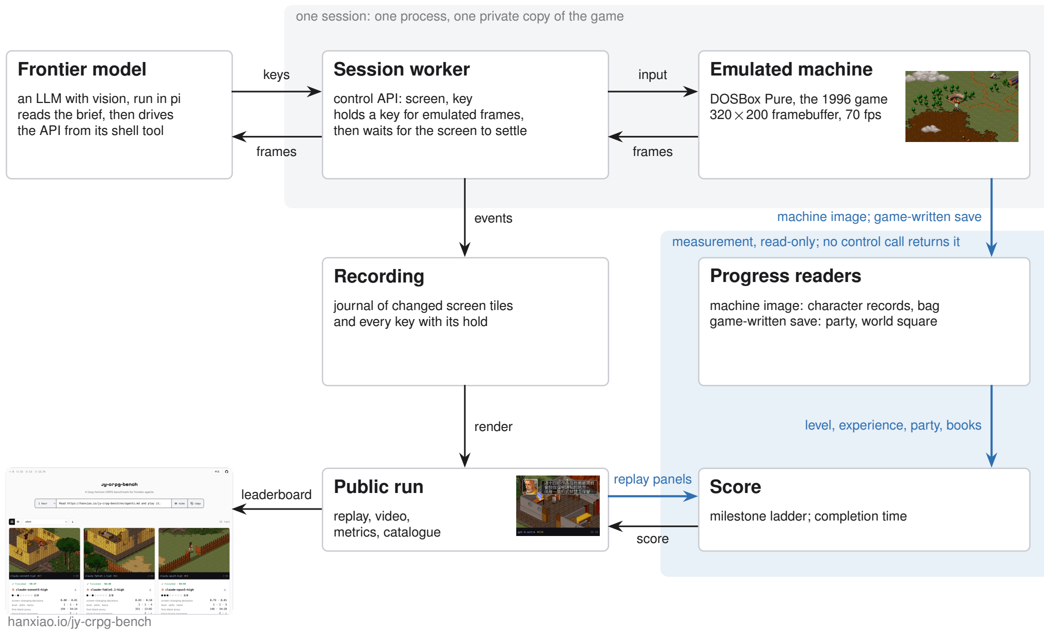

Figure 3: A frontier model, hosted in pi, drives the control API of its own session; the recording feeds the public replay, and progress is read from the emulated machine, from the save the game writes and from the replay, on paths no control call reaches.

For the party roster and the world position the benchmark reads the save the game writes. The game offers its save menu on the world map alone, and the menu itself shows where the party stands, since it has six rows on the world map and four in a scene. An attempt opens the menu, reads its height, and saves only where six rows appear. An attempt recurs every two minutes and once more before the budget lapses, and waits for a two-second gap in the keys the model sends. The session is held while the attempt runs, so the model never sees the screen mid-save, and the time of the first save that lands is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface, so a save that lands is itself the evidence that the party has crossed onto the world map. The session record also keeps a fingerprint of the world map taken from the frame, which credits the crossing only for a session with no save on record.

Five events leave no trace in either record, because the game keeps them only on screen. The model speaks with the hermit, opens the item screen with the compass in it, enters a fight, loses it, or answers the prompt of a companion who asks to join. These are read from the published replay. Each event draws a panel at a fixed position of the screen. The hermit has a portrait in the dialogue frame, and the compass adds a coordinate line to the item screen. A fight shows the card of the acting character, a defeat shows its banner (戰鬥失敗), and a recruitable character asks with a yes-or-no prompt. The panels are fixed art at fixed places, and every frame of the replay is matched against them at one frame a second, with a match above 0.9 counted as a hit. A prompt followed by a yes key in the keypress journal is a recruitment.

Progress is summarised as 10 milestones, in the spirit of the achievements of Crafter (Hafner, 2022). The milestones fall into two phases that the game itself separates. The opening needs no fight. Its five milestones are the chest in the first room, the exit tile, the conversation with the hermit, the compass in his cabinet, and the companion who joins through dialogue, all reached by walking, reading and answering. The campaign begins with the first fight, which a party at level one can lose, and runs through the experience and levels that victories pay and the chains of items, factions and fights that the books sit behind. Its five milestones are a fight entered, a fight fought to its end, experience gained, the second level, and the first of the fourteen books. The default budget is sized to the opening and the extended budgets to the campaign. The milestones are listed in the order of the opening route, but a run is credited with every milestone it reaches, so the count does not depend on the path taken.

The milestones run to the ending of the game. The fourteen books, named in Appendix C, are what the game requires before its final sequence can be played, and the archive records every book held. The count of books therefore extends the milestones beyond the last of them, and the first read that finds all fourteen held records the completion time. A session that completes carries that time, which orders a field in which every entrant passes. For multi-hour budgets the release also records a finer trajectory over the same ground truth: the scenes reached, battles survived, skills learned and books gathered. A long run holding no book therefore still leaves an ordered record of how far its plan carried.

4 EVALUATION

The field is 12 models from 5 vendors at the 60-minute budget, every one run in pi under the same settings, with 42 sessions on record and 2 to 6 per model. A model is credited with every milestone any of its sessions reached, and a session created under a variant spelling of a model name is listed under the model. The floor is a seeded random policy that plays the same budget through the same control surface, drawing from 13 keys the brief teaches, weighted towards movement and confirmation, and never reading the screen (Mnih et al., 2015; Hafner, 2022; ARC Prize Foundation, 2026). Because it is seeded, its key sequence is checked offline against the journal of keys the machine received. Every session is public with its video and replay timeline. The subsections that follow report how many milestones are open, where they separate the models, and how the budget is spent.

4.1 HEADROOM

Of the 12 models, 9 pick something up, all 12 reach the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion, enters a fight and fights it to the end. It reaches 7 of the 10 milestones across its 4 sessions. No model gains experience: every one of the 37 character records read remains at level 1 with 0 experience and 1 skill, as it was when the game began, so every milestone from experience on is open. Figure 4 shows the milestones by model. Comparable benchmarks report their fields at a similar point: BALROG places its best model at 1.5% progression on NetHack (Paglieri et al., 2025), and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025a). The distance from the first fight to the fourteen books is the headroom the benchmark holds.

4.2 DISCRIMINATION

The chest, the hermit and the fight divide the field, each differently, and Figure 5 collects one frame from each pattern behind the divisions. The random policy reaches 1 of the 10 milestones in 60 minutes: in a small room a random walk searches the chest within the hour. Its recomputed key sequence matches the journal of the machine key for key, and no save lands during its hour, so the world map is the first milestone a random walk does not reach. The chest therefore separates models from one another but not from the floor: 9 models searched it and 3 never did, so the count of milestones summarises what a model achieved and presumes no single path.

Every model crosses onto the world map, and the replays show what the crossing costs. The compound is left through a yard closed by a fence with one gate, and the yard held several sessions. `glm-5.3-flash` needed 1595 actions to cross in one session and reached the world map at minute 37 after long stretches pressed into the fence, against a median crossing of 86 actions. The fastest crossing of each model, the bar in Figure 4, runs from 24 actions for `claude-opus5-high` to 130 for `grok-4.6`, with a median of 62. It never pressed enter or space in that session, so the chest in the first room, which opens on either, stayed shut. Beyond the fence the hour goes to the map itself. `claude-fable5.1-high` crossed at minute 13 and then spent 1261 actions and 44 minutes on the world map without entering a scene, and passed the gate of the hermit repeatedly. The entrance of a walled compound is one tile on one face, and every other face is solid. Two sessions that found an inn and a shop talked to the keeper repeatedly, and nothing in the reply advances the opening. `gpt-5.6-terra` sent lists of up to 100 keys in one action, 22 on average, so it looked at the screen once for every 22 keys it pressed, and it reached the world map credited by the save alone. The world-map milestone also shows why the save is read where the game writes one. Of the

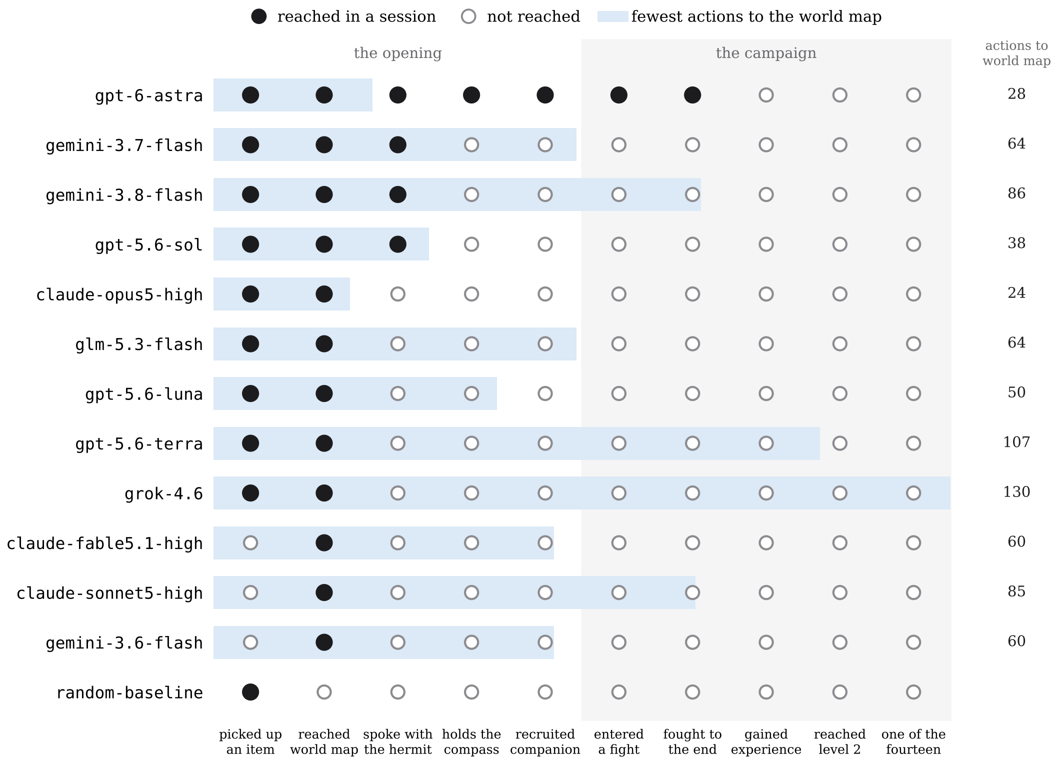

Figure 4: The milestones, one row per model; a filled marker is a milestone reached in at least one of its sessions, and the bar behind the row is the fewest actions any of its sessions took to reach the world map, on a linear scale with the count at the right. The evidence behind each milestone, and the record it is read from: an item count above its opening count (bag in memory); the save the game writes on the world map, or the world-map fingerprint (save, frame); the portrait of the hermit (replay); the compass in the bag or on the item screen (bag, replay); a party of more than one, or the prompt of a companion answered yes (save, replay); the card of the acting character (replay); the defeat banner or experience gained (replay, record); experience above zero and a level above one (record); a book in the bag or on the protagonist (bag, record).

10 crossings that carry both a save and a screen fingerprint, the two agree on 9, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it.

The hermit divides the field a second time, and he divides it on reading. This milestone and the fights and recruitment above it are read from the replay, and no frame without the panel scored above 0.79 against the threshold of 0.9. `gemini-3.7-flash`, `gemini-3.8-flash`, `gpt-5.6-sol` and `gpt-6-astra` reach his conversation, in 8 sessions, and the conversation is long. `gemini-3.8-flash` pressed through his 40 lines at 23 seconds each, so it took 15 minutes and ended at minute 54, when the hermit had named the cabinet and 6 minutes of budget remained. `gemini-3.7-flash` reached him at minute 37 and pressed 60 times in 1 minute. Neither searched the cabinet. The 4 sessions of `gpt-6-astra` all reached him between minutes 11 and 19. The session with the fullest record pressed through the conversation in 7 minutes at 10 seconds a line and searched the cabinet. It then opened the item screen 9 times to read its coordinates from the compass, the one instrument the brief recommends. Its 42 arrow-key presses all fell inside menus and prompts, never on the map, and it entered 11 scenes in the hour.

Above the compass the field is a single model. All 4 sessions of `gpt-6-astra` read the compass between minutes 18 and 29, and from there they diverged. One met Duan Yu (段譽) at an inn at minute 46, answered yes at minute 52 when he asked to travel together, and was handed the manual of his art (六脈神劍譜). One entered a fight at minute 27 and lost it at minute 28, then opened the system menu (系統), loaded the save the benchmark had written, and by minute 34 held the compass again. One entered a fight at minute 40 and sent its last key at minute 42 with the fight still open.

(a) `gpt-6-astra` (b) `gpt-6-astra` (c) `gpt-6-astra`

(d) `gpt-6-astra` (e) `gemini-3.8-flash` (f) `glm-5.3-flash`

(g) `claude-fable5.1-high` (h) `grok-4.6` (i) `gpt-6-astra`

Figure 5: Frames from the published replays, each with the strip of its video naming the model, the action count and the clock: (a) the coordinates of the compass on the item screen; (b) the conversation of the hermit; (c) the prompt of Duan Yu in an inn; (d) a fight; (e) the hermit naming the cabinet; (f) the yard behind the fence; (g) passing the gate of the hermit; (h) talking to an innkeeper; (i) the defeat banner.

The fight divides the field a third time, and there the field ends: the milestones of the campaign beyond it are undivided.

4.3 Budget and Effort

The crossing onto the world map is timed by the game, since the first save it writes there trails the crossing by at most the two-minute cadence of the attempts. By that clock the 10 crossings whose save carries a time fall between 9 and 41 minutes into the session, with a median of 26. One session of `gpt-6-astra` crossed at 9 minutes by that clock, after 33 actions, and spent the rest of the hour beyond the exit. The hour therefore covers the first room and, for one model, the walk south to the cabinet and beyond. The budget is wall clock, so the compute a model spends on a decision is charged against the time it has left to play, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The sessions of the field took between 31 and 1994 actions, and the same conversation with the hermit cost one model 10 seconds a line and another 23. A comparison of token spend across models waits for a field in which more than one model plays past the opening. A model that has not reached the hermit has not begun the game the milestones score, and at this budget `gpt-6-astra` alone has, so a cost per milestone would rank one model.

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark built on a 1996 open-world role-playing game that asks a frontier model for its whole repertoire at once. The model must perceive isometric pixel art and Traditional Chinese, bring knowledge of the novels the world is built from, and act through a keyboard alone. It must keep a world model of its own, plan without a quest log or score, and fight turn-based battles against the AI of the game, over thousands of dependent steps with sparse reward. Every model plays under the same brief and control surface and is scored against events of the game itself. At the 60-minute budget the field of 12 models stays inside the opening: every model reaches the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion and fights, while none gains experience. Every milestone from experience to the fourteenth book stands open, and the one milestone the random baseline reaches marks the floor a model has to clear. That a single frontier model plays the game at all, and that even it stands at its first fight with fourteen books still to find, is the measure of the task. The ending is tens of hours of coherent, self-directed play away. The milestones run to that ending, and from the first finish the leaderboard ranks by completion time, with the video and every key of the run on record. Three fields follow this one. Sessions of four to twenty-four hours bring the campaign milestones and a completion time within reach and make a token cost per milestone a comparison between models. The same models with web access measure what a walkthrough is worth against the brief alone. Once a model finishes, the completion clock defines the speedrun.

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A Session and Infrastructure Parameters

A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only. The published video runs at 8 times speed, or faster for long runs, with a strip naming the actor, the action number, the keys held and the clock; a timeline file beside it carries every keypress and its hold.

B Release and Licensing

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified. The bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright Soft-World International (智冠科技) and Heluo Studio. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The frames of the game in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

C The Fourteen Books

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).